

Set data type:

Set is a collection of homogenous data item or heterogeneous data item which is enclosed between the pair of flower brackets.

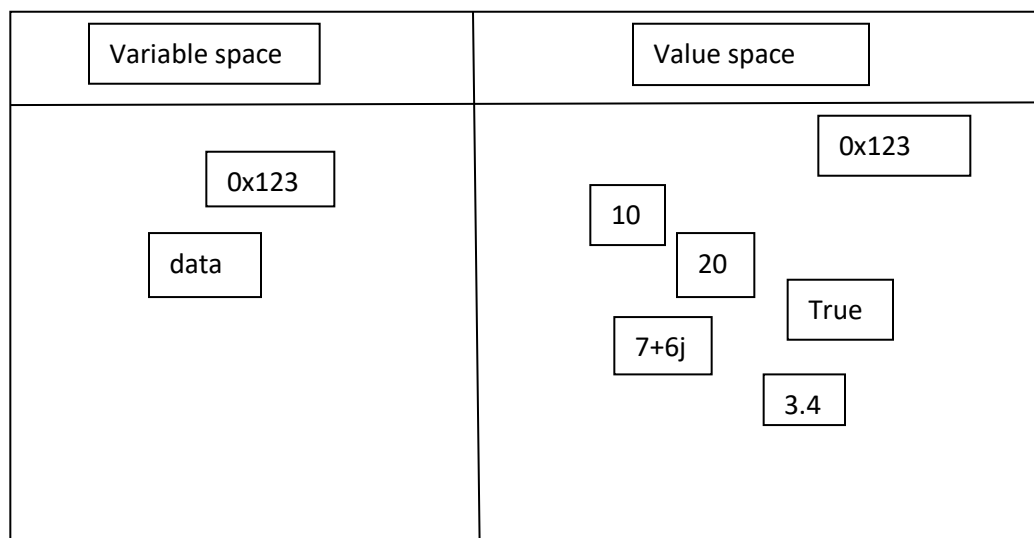
Syntax:

```
Var = {val1, val2, val3.....}
```

- The values should be immutable i.e.,(int, float, complex, string, tuple).
- Set is unordered in nature (the values will not be stored in a same order like how the user have given).
- In set duplicated values will get eliminated.
- Indexing is not possible in set collection because of its unordered nature.

Memory allocation in set data type:

```
data = {True, 10, 20, 3.4, 7+6j, 1, 10}
```



- It is not possible to access values from set using indexing and not possible to modify values based on indexing.

Attributes on set data type:

- **Add ()**: It is a function which is used to add a new value to the existing set collection.
`Variable.add(value)`
- **Remove ()**: It is a function which is used to remove an existing value from the given set collection.
`Variable.remove(value)`
- **Update ()**: It update the current set by adding items from any iterable.
`Variable.update(iterable)`
- **Discard ()**: It is a function which is used to remove specified item from the set.
`Variable.discard(item)`
- **Pop ()**: It is a function which is used to remove random item from the set. It will return removed item.
`Variable.pop()`
- **Clear ()**: It is a function which is used to remove the all the item from the set.
`Variable.clear()`
- **Isdisjoint ()**: It returns True if none of the item present in both the set or else False.
`Base_set.isdisjoint(sub-set)`
- **Issubset ()**: It returns True if all the item from the base set exists in the reference set, or else False.
`Base_set.issubset(sub-set)`
- **Issuperset ()**: It returns True if all the items in the reference set exist in the base set.

Base_set.issubset(sub-set)

- **Union ()**: It returns a set that contains all the items from the original set and all items from specified set.

Set1.union(set2)

- **Intersection ()**: It returns set that contains the similarity between two or more sets.

Set1.intersection(set2)

- **Difference ()**: It returns a set containing the difference between two or more sets. The returned set contains items that exist only in base set.

-base set: the first set which we write before applying function.

-reference set /subset: the set which is written inside the round brackets.

Set1.difference(set2)

- **Symmetric_difference ()**: It returns a set which contains all the items which are not common among all the set.

Set1.Symmetric_difference (set2)

- **Intersection_update ()**: It modifies the set by retaining the only elements found in both set.

Set1.Intersection_update(set2)

- **Difference_update ()**: It update the base set with the elements that are present in base set but not in other set.

Set1.difference_update(set2)

- **Symmetric_difference_update ()**: It updates the base items that are not common among all the sets (both the set)

Set1.Symmetric_difference_update (set2)

Note:

- We can modify the set collection so it is a mutable data type.
- Since set will not remove duplicate values, it is used in “data filtering process”.